MILITARY SPECIFICATION

CYLINDER FOR M56 MINES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the inspection and test of a cylinder used to contain XM56 Anti-Tank Mines.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification For

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment for Inert Ammunition Components.
MIL-STD-1235 - Single and Multilevel Continuous Sampling.

FSC: 1345

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DRAWINGS

U.S. ARMY PICATINNY ARSENAL

PRODUCT DRAWING

9287690 - Cylinder

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. - The assembly shall comply with all requirements specified on Drawing (dwg. 9255963 and with all requirements specified on applicable specifications.

3.3 Hydrostatic Pressure Test

3.3.1 Low Pressure. - The cylinder shall not rupture, leak or deform when subjected to an internal hydrostatic pressure of fifty plus ten (50 + 10) pounds per square inch gage (psig) for a period of five plus one (5 + 1) seconds.

3.3.2 High Pressure. - The cylinder shall not rupture, leak or deform when subjected to an internal hydrostatic pressure of three-hundred plus ten (300 + 10) psig for a period of five plus one (5 + 1) seconds.

3.4 Scratches or extrusion marks.

3.4.1 Scratches or extrusion marks on inside diameter of 0.0015 inches deep, maximum (max.) and 1/16 inch in width, max. will not be cause for rejection.

3.4.2 Scratches or extrusion marks on outside diameter 0.0015 inch deep max. and 1/16 inch in width, max. will not be cause for rejection.

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3.4.3 Scratches or extrusion marks under 0.0015 inch deep max. and 1/16 inch wide max. running the full length of the outside diameter will not be cause for rejection.

3.5 Mechanical Properties. - Cold-working or heat-treating (including quenching-and-tempering, normalizing, normalizing-and-tempering, annealing, and stress-relieving) operations, shall be performed in such a manner as to produce the specified properties uniformly throughout the entire lot. The contractor shall determine the detailed procedures to produce parts meeting the required mechanical properties. These procedures shall be documented and submitted with the first article sample to the Contracting Officer. Changes in these procedures shall also be documented and submitted to the Contracting Officer.

3.6 Workmanship. - All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter, surface cracks. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

3.7 Submission of results of contractor-conducted examinations and tests. - The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract (see 6.4).

3.8 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such

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inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of ten (10) cylinders. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of con-

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formance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as dtermined by the Government.

4.2.2 Inspections to be Performed. - First Article Tests and inspections shall be performed to the following requirements.

a. Ten (10) cylinders will be subjected to the applicable examinations of 4.3.2.1.

b. Ten (10) cylinders shall be sequentially tested for:

1) Hardness test (see 4.4.1)

2) Hydrostatic Test (see 4.4.2 and 4.4.3)

c. Four (4) cyclinders shall be selected from 4.2.2.b and subjected to the salt spray test in accordance with 4.3.3.1.

4.2.3 Rejection. - If any cylinder fails to comply with the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon encountering any failure of an item to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the

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homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cylinder (see dwg. 9287690).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Pitch diameter of cartridge threads	Gage	01001
Major: AQL 1.5 percent			
101.	Minimum length of cartridge thread	Gage	01002
102.	Depth of pressure well	Gage	01003
103.	Distance from top of cylinder to inside surface of cylinder containing propagation hole ..	Gage	01004
104.	Overall outside length from bottom opening to shoulder ...	Gage	01005
105.	Location of helix angle basic points on lug	Gage	01006

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106.	Outside diameter (in restrained condition in accordance with Note 8 of the drawing)	Gage	01007
107.	Location of top diameter	Gage	01008
108.	Lugs diameter	Gage	01009
109.	Lugs diameter location	Gage	01010
110.	Cartridge thread location	Gage	01011
111.	Propagation hole diameter	Gage	01012
112.	Perpendicularity of top surface of cylinder large diameter to top surface of cap ...	Gage	01013
113.	Distance from inside of cylinder at minimum thickness to top surface of cylinder large diameter	Gage	01014
114.	Inside diameter of large diameter (in restrained condition and in accordance with Note 7 of the drawing)	Gage	01015
Minor:	AQL 4.0 percent		
201.	Number of teeth	Visual	01016
202.	Scratches or extrusion marks (see 3.4)	Gage	01017
203.	Location of propagation hole .	Gage	01018
204.	Gas chamber diameter	Gage	01019
205.	Burrs or high spots on chamfers inside of large diameter.	Gage/ Visual	01020
206.	Location of wrench slots	Gage	01021
207.	Overall length	Gage	01022
208.	Evidence of poor workmanship (see 3.6)	Visual	01023

4.3.3 TESTING

4.3.3.1 Salt Spray, (see dwg. 9287690) Major Defect Code No. 02001. - Four (4) cylinders shall be randomly selected from each lot and subjected to the salt spray test requirements specified on the applicable drawing. Testing shall be in accordance with paragraph 811.1 of FED-STD-151. Any cylinder failing to comply with all of the specified requirements shall be classed defective. The lot shall be rejected if one (1) or more cylinders are classed defective. Testing is destructive and samples shall not be returned to the lot.

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4.3.3.2 Hardness Test (see dwg. 9287690), Major Defect, Code No. 02002.- Three hundred and fifteen (315) cylinders shall be randomly selected from each lot and tested in accordance with 4.4.1. The lot shall be rejected if three (3) or more cylinders are classed defective.

4.3.3.3 Hydrostatic Low Pressure Test. - (see 3.3.1) Major Defect, Code No. 02004. - Three hundred and fifteen (315) cylinders shall be tested in accordance with 4.4.2. The lot shall be rejected if one (1) or more cylinders are classed defective (Non-destructive test).

4.3.3.4 Hydrostatic High Pressure Test. - (see 3.3.2), Major Defect, Code No. 02005. - Twenty five (25) cylinders that have successfully passed the hydrostatic low pressure test of 4.3.3.3 shall be tested in accordance with 4.4.3. The lot shall be rejected if one (1) or more cylinders are classed defective (Destructive test)

4.3.4 Inspection Equipment. - Inspection Equipment required to perform the isnpections prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection design and approval.

4.4 TEST METHODS AND PROCEDURES

4.4.1 Hardness Test. - The cylinder shall be subjected to the hardness test after application of the protective finish and in accordance to the procedure specified in Method 243.1 of FED-STD-151. Three (3) readings shall be taken on the cylinder; one (1) reading shall be on the tube, one (1) on the insert, and one (1) on the cap (on the lugs). The cylinder shall be classed defective if it fails to comply with all of the requirements. Testing is non-destructive and only accepted cylinders shall be returned to the lot.

4.4.2 Hydrostatic Test. (Low Pressure). - The cylinder shall be mounted in an approved fixture and the specified pressure shall be applied for the required time. The cylinder shall be classed defective if it fails to comply with all of the specified requirements. Testing is non-destructive; samples may be returned to the lot.

4.4.3 Hydrostatic Test (High Pressure). - The cylinder shall be mounted in an approved fixture and the specified pressure shall be applied for the required time. The cylinder shall be classed defective if it fails to comply with all of the specified requirements. Testing is destructive; samples shall not be returned to the lot.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C. - The cylinders shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing

5.2.1 Level C. - The cylinders preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. - Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping. - When cylinders from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The

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contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801

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in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of results of contractor conducted examinations and tests (see 3.7). - The contractor shall send his results to:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-M
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1345-A-039

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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